

# Task 9 — Trend Forecasting

Data Analyst · Phase 1 Industry Immersion · Study Guide

## DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Building · Proficient

Level 9 of 20 · ramp 43%

### 1 · What this is & why it matters

Forecast key trends forward using historical patterns so the business can plan, not just react.

Anticipating demand, churn or load turns analytics from a rear-view mirror into a planning tool.

### 2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A forecast with validation error vs a baseline and clearly shown uncertainty.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

### 3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

### 4 · Key concepts (understand these first)

#### Trend vs seasonality vs noise

the components of a time series.

#### Train/validation split for time

never shuffle; the future must be held out.

#### Forecast horizon

how far ahead you predict, and growing uncertainty with distance.

#### Error metrics

MAE/MAPE to judge forecast quality honestly.

#### Confidence intervals

a range, not a single false-precise number.

### 5 · Recommended stack & tools

- pandas
- statsmodels / Prophet
- matplotlib

### 6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Visualise the series and identify trend and seasonality.
2. Hold out the most recent period as a validation set.

3. Fit a simple baseline (naive/last-value) before anything fancy.
4. Fit a forecasting model and compare against the baseline by error metric.
5. Produce forecasts with confidence intervals.
6. State assumptions and where the forecast is least reliable.

## 7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Classical (ARIMA/ETS) vs Prophet vs a simple moving average baseline.
- Point forecast vs probabilistic forecast.

## 8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- What event could break this forecast (a launch, a holiday)?
- Is the fancy model actually beating the naive baseline?
- How will you show uncertainty without overwhelming readers?

## 9 · Definition of Done

### Done when all are true and demoable

- A forecast with validation error vs a baseline and clearly shown uncertainty.
- Demonstrable live on real data.

## 10 · How your work is evaluated (out of 100)

| Scoring parameter                                                                                | Marks      |
|--------------------------------------------------------------------------------------------------|------------|
| Core deliverable — A forecast with validation error vs a baseline and clearly shown uncertainty. | 50         |
| Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)           | 20         |
| Live verification & evidence (demonstrated live; real output, not claims)                        | 15         |
| Dependency, failure & edge-case handling (errors handled; hand-offs honoured)                    | 15         |
| <b>TOTAL</b>                                                                                     | <b>100</b> |

## 11 · Pitfalls to avoid

### Worry if any of these are true of your work

- Shuffling time data and leaking the future.
- No baseline to beat.
- Presenting a single number as if it were certain.

## 12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

### Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design
- Time-series and segmentation techniques